

Linda Miao

Quiz 2:

-- This quiz has 7 questions. Please attempt all questions. Take screenshots of the query plan you get add an explanation

-- for each query plan. Put the screenshots of the query plans and the explanation in one PDF file and upload the PDF file

-- Please uncomment these lines in case you want to drop and recreate the QO_Fun_Quiz1

--USE master

--GO

--DROP DATABASE QO_Fun_Quiz1

--GO

CREATE DATABASE QO_Fun_Quiz1

GO

USE QO_Fun_Quiz1

GO

-- Creating 3 tables T1, T2, T3

CREATE TABLE T1

(

 a int not null,

 b int

)

```
CREATE TABLE T2
```

```
(  
    a int not null,  
    b int  
)
```

```
CREATE TABLE T3
```

```
(  
    a int not null,  
    b int not null  
)
```

```
GO
```

```
-- adding rows to the tables 1, T2 and T3
```

```
DECLARE @i1 int = 0
```

```
WHILE @i1 < 10000
```

```
BEGIN
```

```
    SET @i1 = @i1 + 1
```

```
    INSERT INTO T1 values(@i1, @i1*10)
```

```
END
```

```
DECLARE @i2 int = 0
```

```
WHILE @i2 < 30000
```

```
BEGIN
```

```
    SET @i2 = @i2 + 1
```

```
INSERT INTO T2 values(@i2, @i2*10)

END
```

```
DECLARE @i3 int = 0

WHILE @i3 < 70000

BEGIN

    SET @i3 = @i3 + 1

    INSERT INTO T3 values(@i3, @i3*10)

END

GO
```

-- Please consider the following query for the following questions

```
SELECT *

FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b
```

-- Q1

-- Q2

alter table T1 add CONSTRAINT T1_PK primary key (a)

Q1, No indexes: I ran the SELECT with no indexes on any table. The query plan showed Hash Match joins. This is because T1, T2, and T3 have no indexes and data is not sorted. The optimizer picked Hash Match because it works best for large unsorted tables. T1 is the smallest table (5,000 rows) so it was used as the build side to save memory.

SQLQuery1.s...aptop (55))*

```

33  SET @i3 = @i3 + 1
34  INSERT INTO T3 values(@i3, @i3*10)
35  END
36
37  -- q1: enable plan and select run
38  SELECT *
39  FROM T1 JOIN T2 on T1.a = T2.a
40  JOIN T3 on T1.a = T3.b

```

100 % No issues found Ln: 38, Ch: 1 (67 chars, 3 lines) SPC CRLF UTF-8

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

SQLQuery1.s...aptop (55))*

```

33  SET @i3 = @i3 + 1
34  INSERT INTO T3 values(@i3, @i3*10)
35  END
36
37  -- q1: enable plan and select run
38  SELECT *
39  FROM T1 JOIN T2 on T1.a = T2.a
40  JOIN T3 on T1.a = T3.b

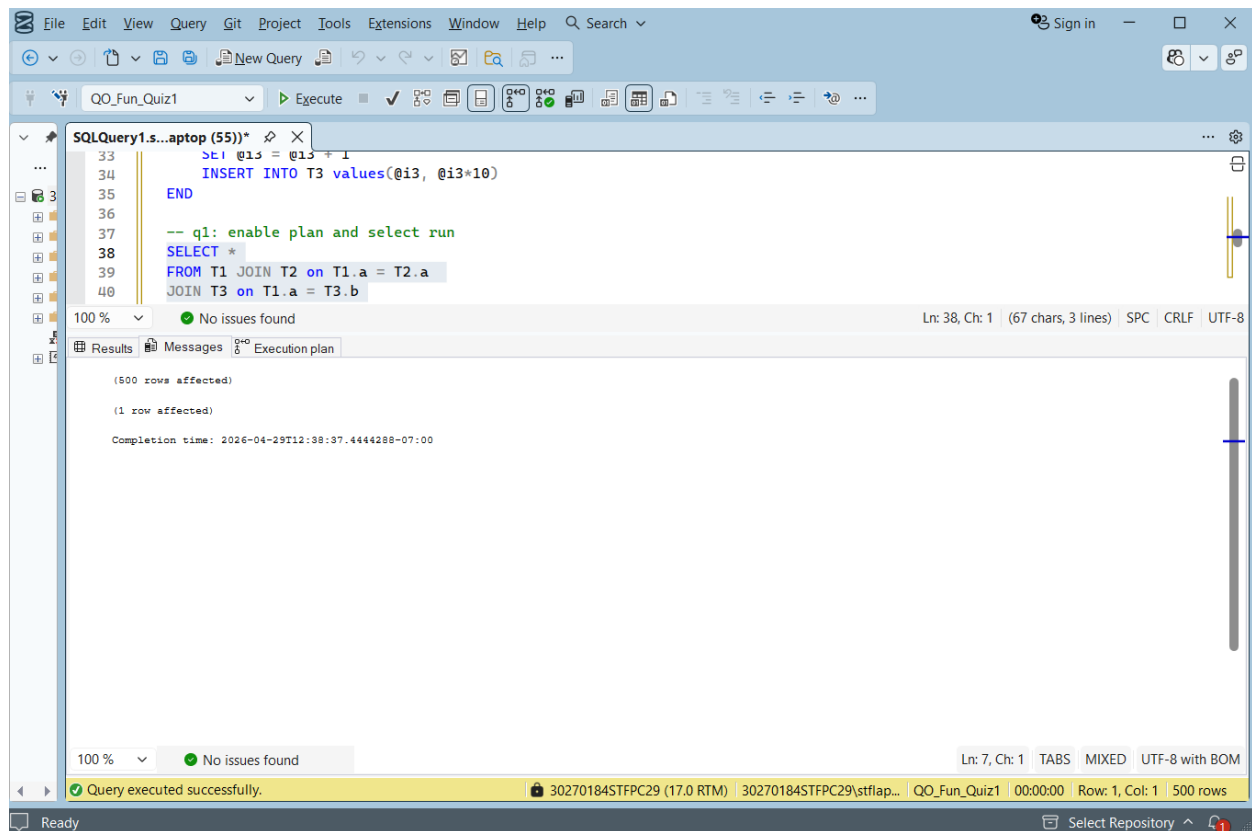
```

100 % No issues found Ln: 38, Ch: 1 (67 chars, 3 lines) SPC CRLF UTF-8

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

SELECT	
Cached plan size	72 KB
Estimated Operator Cost	0 (0%)
Degree of Parallelism	1
Estimated Subtree Cost	0.196421
Memory Grant	2752 KB
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	499.155
Statement	
SELECT *	
FROM T1 JOIN T2 on T1.a = T2.a	
JOIN T3 on T1.a = T3.b	



alter table T2 add CONSTRAINT T2_PK primary key (a)

Q2 , Clustered index on T1(a) and T2(a): After adding primary keys on T1(a) and T2(a), the query plan shows Hash Match joins with Clustered Index Scans on T1 and T2. With only 5,000 rows in each table, the optimizer chose Hash Match as the most cost efficient plan. The clustered indexes are visible as Clustered Index Scan operators in the plan.

File Edit View Query Git Project Tools Extensions Window Help Search

Sign in

QO_Fun_Quiz1

Execute

SQLQuery1.s...aptop (55))

```
32 BEGIN
33     SET @i3 = @i3 + 1
34     INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)
45 -- q3:
46 alter table T3 add CONSTRAINT T3_PK primary key (a)
47 --q4:
48 alter table T3 drop constraint T3_PK
49 alter table T3 add CONSTRAINT T3_PK primary key (b)
```

100 % No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

(500 rows affected)

(1 row affected)

Completion time: 2026-04-29T13:02:16.1287589-07:00

100 % No issues found Ln: 7, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready Select Repository

File Edit View Query Git Project Tools Extensions Window Help Search

QO_Fun_Quiz1 Execute

SQLQuery1.s...aptop (55))*

```
32 BEGIN
33 SET @i3 = @i3 + 1
34 INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
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41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)
45 -- q3:
46 alter table T3 add CONSTRAINT T3_PK primary key (a)
47 --q4:
48 alter table T3 drop constraint T3_PK
49 alter table T3 add CONSTRAINT T3_PK primary key (b)
```

100 % No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

Hash Match (Inner Join)
Cost: 25 %
0.022s
500 of 499 (100%)

Hash Match (Inner Join)
Cost: 50 %
0.013s
500 of 499 (100%)

Clustered Index Scan (Cluste...
[T1].[T1_PK]
Cost: 8 %
0.002s
5000 of 5000 (100%)

Clustered Index Scan (Cluste...
[T2].[T2_PK]
Cost: 8 %
0.007s

Clustered Index Scan (Cluste...
[T3].[T3_PK]
Cost: 8 %
0.006s

Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready Select Repository

SQLQuery1.s...aptop (55))*

```

32 BEGIN
33     SET @i3 = @i3 + 1
34     INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)
45 -- q3:
46 alter table T3 add CONSTRAINT T3_PK primary key (a)
47 --q4:
48 alter table T3 drop constraint T3_PK
49 alter table T3 add CONSTRAINT T3_PK primary key (b)

```

100 % No issues found Lr: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%
 SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

SELECT	
Cache plan size	72 KB
Estimated Operator Cost	0 (0%)
Degree of Parallelism	1
Estimated Subtree Cost	0.196421
Memory Grant	2752 KB
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	499.155

Statement

```

SELECT *
FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b

```

30270184STFPC29 (17.0 RTM) | 30270184STFPC29\stflap... QO_Fun_Quiz1 | 00:00:00 | Row: 1, Col: 1 | 500 rows

Ready Select Repository

-- Q3

alter table T3 add CONSTRAINT T3_PK primary key (a)

Q3 Clustered index on T3(a): I added a clustered index on T3(a) but the query plan did not change. This is because the join condition uses T3.b, not T3.a. The index on a does not help when joining on b, so the optimizer ignored it and kept Hash Match for the T3 join.

File Edit View Query Git Project Tools Extensions Window Help Search

Sign in

QO_Fun_Quiz1 Execute

SQLQuery1.s...aptop (55)*

```
32 BEGIN
33 SET @i3 = @i3 + 1
34 INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)
```

100 % No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

```
graph LR
    S[SELECT] --> H1[Hash Match (Inner Join)]
    H1 --> H2[Hash Match (Inner Join)]
    H2 --> C1[Clustered Index Scan (Cluste...)]
    H2 --> C2[Clustered Index Scan (Cluste...)]
    H2 --> C3[Clustered Index Scan (Cluste...)]
```

Operation	Cost	Time	Rows	Percentage
SELECT	0 %			
Hash Match (Inner Join)	25 %	0.016s	500 of 499	100%
Hash Match (Inner Join)	50 %	0.011s	500 of 499	100%
Clustered Index Scan (Cluste...)	8 %	0.001s	5000 of 5000	100%
Clustered Index Scan (Cluste...)	8 %	0.003s	5000 of 5000	100%
Clustered Index Scan (Cluste...)	8 %	0.005s	5000 of 5000	100%

Query executed successfully.

30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready Select Repository

File Edit View Query Git Project Tools Extensions Window Help Search

Sign in

QO_Fun_Quiz1 Execute

SQLQuery1.s...aptop (55)*

```

32 BEGIN
33     SET @i3 = @i3 + 1
34     INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)

```

100% No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%
 SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

Hash Match (Inner Join) Hash Match (Inner Join) Clustered Index Scan (Cluste... [T1].[T1_PK] Cost: 8 % 0.001s 5000 of 5000 (100%)

SELECT

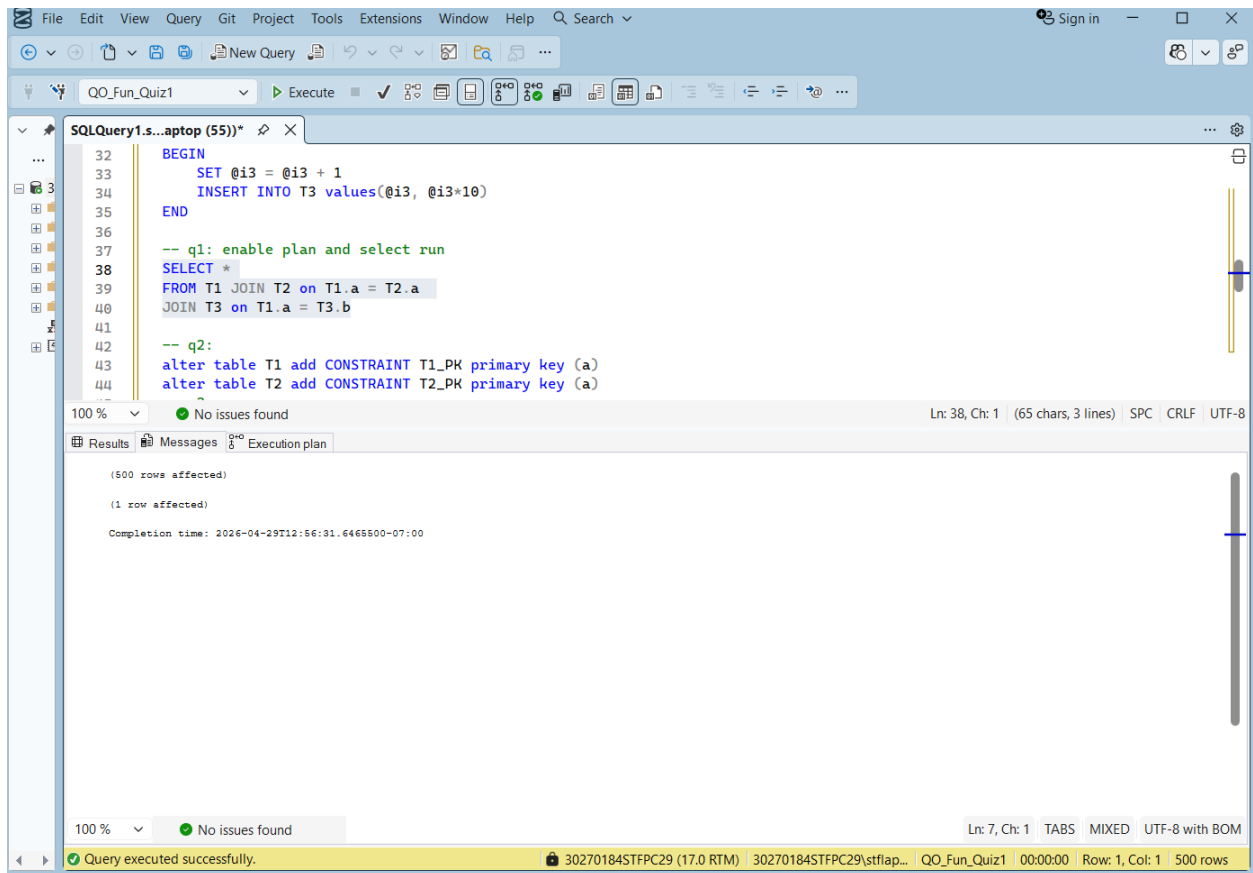
Cached plan size	72 KB
Estimated Operator Cost	0 (0%)
Degree of Parallelism	1
Estimated Subtree Cost	0.196421
Memory Grant	2752 KB
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	499.155

Statement
 SELECT *
 FROM T1 JOIN T2 on T1.a = T2.a
 JOIN T3 on T1.a = T3.b

Clustered Index Scan (Cluste... [T3].[T3_PK] Cost: 8 % 0.005s 5000 of 5000 (100%)

302701845TFPC29 (17.0 RTM) 302701845TFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready Select Repository



-- Q4

alter table T3 drop constraint T3_PK

alter table T3 add CONSTRAINT T3_PK primary key (b)

Q4 Clustered index on T3(b): After dropping T3(a) and creating the index on T3(b), all joins switched to Merge Join. Now T3 is sorted by b which matches the join condition T1.a = T3.b. Since all tables are sorted on their join columns the optimizer chose Merge Join for all joins.

File Edit View Query Git Project Tools Extensions Window Help Search

Sign in

QO_Fun_Quiz1 Execute

SQLQuery1.s...aptop (55))*

```
32 BEGIN
33     SET @i3 = @i3 + 1
34     INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:
43 alter table T1 add CONSTRAINT T1_PK primary key (a)
44 alter table T2 add CONSTRAINT T2_PK primary key (a)
45 -- q3:
46 alter table T3 add CONSTRAINT T3_PK primary key (a)
47 --q4:
48 alter table T3 drop constraint T3_PK
49 alter table T3 add CONSTRAINT T3_PK primary key (b)
50
51 --q5:
52 alter table T3 drop constraint T3_PK
53 alter table T3 add CONSTRAINT T3_PK primary key (a,b)
54
55 --q6:
56 alter table T3 drop constraint T3_PK
```

100 % No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

(500 rows affected)

(1 row affected)

Completion time: 2026-04-29T12:58:15.1178380-07:00

100 % No issues found Ln: 7, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready Select Repository

FileEditViewQueryGitProjectToolsExtensionsWindowHelpSearch

QO_Fun_Quiz1Execute

SQLQuery1.s...aptop (55))*

32BEGIN
33SET @i3 = @i3 + 1
34INSERT INTO T3 values(@i3, @i3*10)
35END
36
37-- q1: enable plan and select run
38SELECT *
39FROM T1 JOIN T2 on T1.a = T2.a
40JOIN T3 on T1.a = T3.b
41
42-- q2:

100 %No issues foundLn: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

ResultsMessagesExecution plan

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

SELECT
Cost: 8 %

Hash Match
(Inner Join)
Cost: 25 %

Hash Match
(Inner Join)
Cost: 50 %

Clustered Index Scan (Cluste...
[T1].[T1_PK]
Cost: 8 %
0.001s
5000 of
5000 (100%)

Clustered Index Scan (Cluste...
[T3].[T3_PK]
Cost: 8 %
0.006s
5000 of
5000 (100%)

Cached plan size72 KB
Estimated Operator Cost0 (0%)
Degree of Parallelism1
Estimated Subtree Cost0.196421
Memory Grant2752 KB
Estimated Number of Rows for All Executions0
Estimated Number of Rows Per Execution499.155

Statement
SELECT *
FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b

Query executed successfully.30270184STFPC29 (17.0 RTM)30270184STFPC29\stflap...QO_Fun_Quiz100:00:00Row: 1, Col: 1500 rows

ReadySelect Repository

SQLQuery1.s...aptop (55)*

```

32 BEGIN
33 SET @i3 = @i3 + 1
34 INSERT INTO T3 values(@i3, @i3*10)
35 END
36
37 -- q1: enable plan and select run
38 SELECT *
39 FROM T1 JOIN T2 on T1.a = T2.a
40 JOIN T3 on T1.a = T3.b
41
42 -- q2:

```

100 % No issues found Ln: 38, Ch: 1 (65 chars, 3 lines) SPC CRLF UTF-8

Query 1: Query cost (relative to the batch): 100%
SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

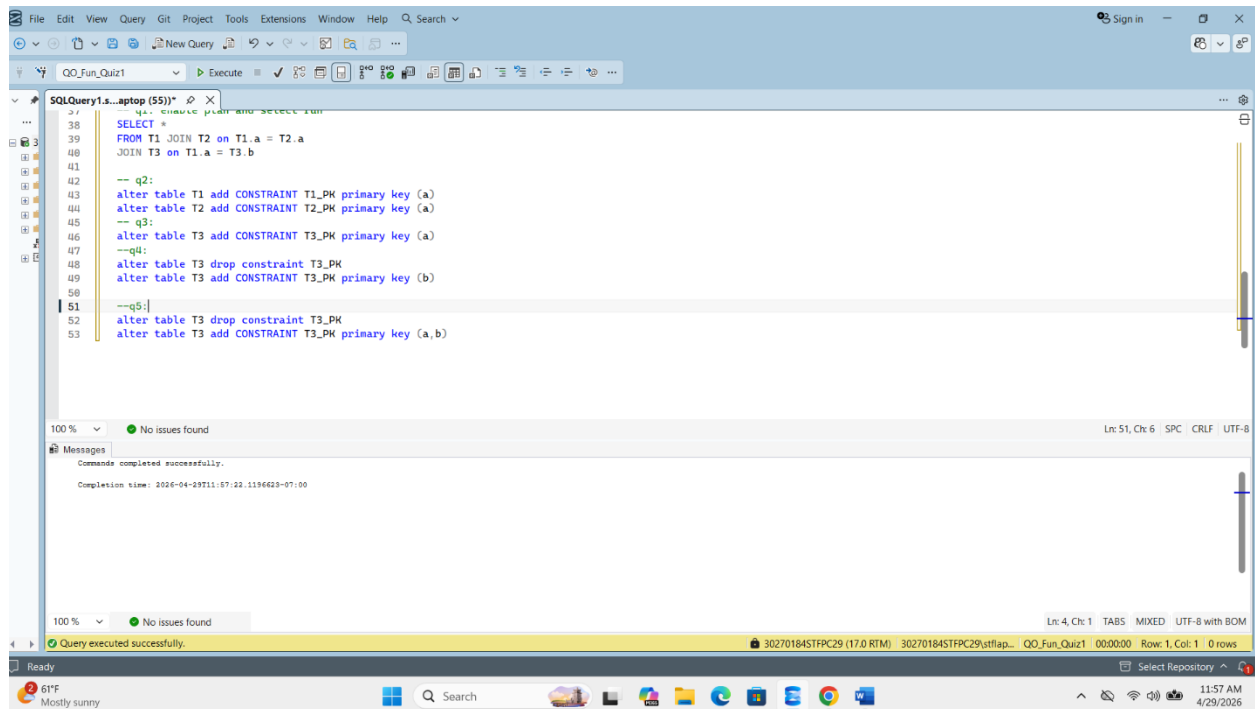
Query executed successfully. 30270184STFPC29 (17.0 RTM) 30270184STFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

-- Q5

alter table T3 drop constraint T3_PK

alter table T3 add CONSTRAINT T3_PK primary key (a,b)

Q5 Clustered index on T3(a,b): After changing the index to (a,b), the plan reverted back to Hash Match for the T3 join. Even though b is in the index, a is the leading column so the data is sorted by a first. This means T3 is not globally sorted by b alone, so the optimizer cannot use Merge Join on b and switched back to Hash Match.



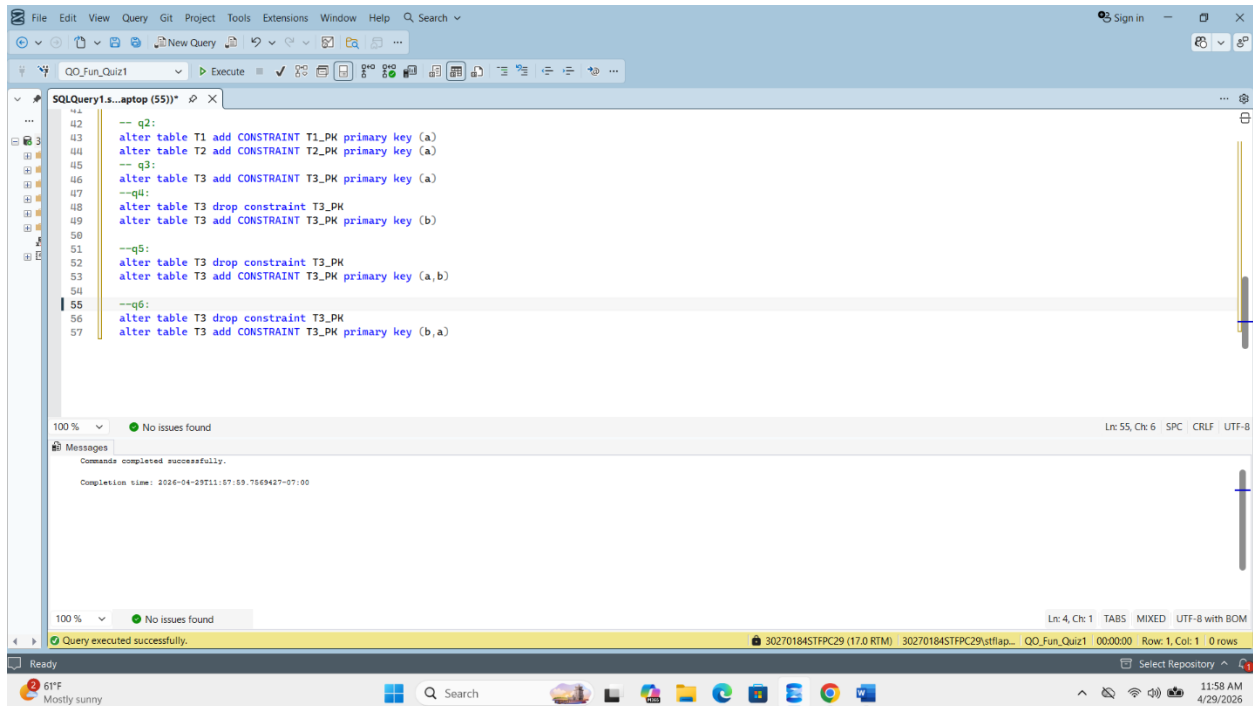
-- Q6

alter table T3 drop constraint T3_PK

alter table T3 add CONSTRAINT T3_PK primary key (b,a)

Q6 explanation: After creating index T3(b,a), the plan shows Hash Match. With 5,000 rows the optimizer chose Hash Match as the most cost efficient plan. T3 is sorted by b as the

leading column which matches the join condition T1.a = T3.b.



```
-- q2:
alter table T1 add CONSTRAINT T1_PK primary key (a)
alter table T2 add CONSTRAINT T2_PK primary key (a)
-- q3:
alter table T3 add CONSTRAINT T3_PK primary key (a)
--q4:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b)
--q5:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (a,b)
--q6:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b,a)
```

100 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-04-29T11:57:59.7569427-07:00

100 % No issues found

Ln: 55, Ch: 6 SPC CRLF UTF-8

Ln: 4, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully.

302701845TPC29 (17.0 RTM) 302701845TPC29,utflap... QO_Fun_Quiz1 000000 Row: 1, Col: 1 0 rows

Ready 61°F Mostly sunny 11:58 AM 4/29/2026

-- Q7

SELECT *

FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b

Where T1.a>@param

Q7 — Different @param values:

- T1.a > 9999 — T1 returns 0 rows since all values are 1 to 5000. The optimizer chose Nested Loop because joining 0 rows costs almost nothing.
- T1.a > 0 — T1 returns all 5000 rows. The optimizer used Merge Join or Hash Match since all rows participate.
- T1.a > 2500 — T1 returns about half the rows. The optimizer may choose Nested Loop or Merge Join depending on the estimated row count and cost.

SQLQuery1.s...aptop (55)*

```
--q6:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b,a)

--q7:
-- Test 1: large value (returns 0 rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
```

100 % No issues found

Results Messages Execution plan

(0 rows affected)

(1 row affected)

Completion time: 2026-04-29T11:59:55.9906614-07:00

Ln: 61, Ch: 1 (82 chars, 2 lines) SPC CRLF UTF-8

Query executed successfully.

SQLQuery1.s...aptop (55)*

```
--q6:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b,a)

--q7:
-- Test 1: large value (returns 0 rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
```

100 % No issues found

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 9999

Execution plan diagram showing Nested Loops (Inner Join) and Clustered Index Seek operations.

Query executed successfully.

File Edit View Query Git Project Tools Extensions Window Help Search

SQLQuery1.s...aptop (SS) * X

```
--q6:
55 alter table T3 drop constraint T3_PK
56 alter table T3 add CONSTRAINT T3_PK primary key (b,a)
57
58
59 --q7:
60 -- Test 1: large value (returns 0 rows)
61 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
62 JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
63
```

100% No issues found Lrn 61, Ch 1 (82 chars, 2 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 9999

SELECT

Cache plan size 40 KB

Estimated Operator Cost 0 (0%)

Degree of Parallelism 1

Estimated Subtree Cost 0.0098586

Estimated Number of Rows for All Executions 0

Estimated Number of Rows Per Execution 1

Statement

SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 9999

Clustered Index Seek (Cluste...
[T1].[T1_PK]
Cost: 33 %
0.000a
0 of
1 (0%)

Clustered Index Seek (Cluste...
[T2].[T2_PK]
Cost: 33 %
0.000a
0 of
1 (0%)

Clustered Index Seek (Cluste...
[T3].[T3_PK]
Cost: 33 %
0.000a
0 of
1 (0%)

Query executed successfully. 302701845TFPC29 (17.0 RTM) 302701845TFPC29\stflap... QO_Fun_Quiz1 00:00:00 Row: 1, Col: 1 0 rows

Ready 61°F Mostly sunny Search 12:00 PM 4/29/2026

File Edit View Query Git Project Tools Extensions Window Help Search

SQLQuery1.s...aptop (55) X

```
--q6:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b,a)

--q7:
-- Test 1: large value (returns 0 rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 9999

-- Test 2: small value (returns all rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 0

-- Test 3: middle value
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 2500
```

100 % No issues found Lnt: 64, Ch: 1 (124 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

(500 rows affected)

(1 row affected)

Completion time: 2026-04-29T12:01:43.9129943-07:00

100 % No issues found Lnt: 7, Ch: 1 TABS MIXED UTF-8 with BOM

Query executed successfully. 302701845TFPC29 (17.0 RTM) 302701845TFPC29;stflap... QQ_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

File Edit View Query Git Project Tools Extensions Window Help Search

SQLQuery1.s...aptop (55) X

```
--q6:
alter table T3 drop constraint T3_PK
alter table T3 add CONSTRAINT T3_PK primary key (b,a)

--q7:
-- Test 1: large value (returns 0 rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 9999

-- Test 2: small value (returns all rows)
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 0

-- Test 3: middle value
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 2500
```

100 % No issues found Lnt: 64, Ch: 1 (124 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 0

Hash Match (Inner Join) Cost: 25 % 500 of 499 (100%)

Hash Match (Inner Join) Cost: 50 % 500 of 499 (100%)

Clustered Index Seek (Cluste... [T1].[T1_PK] Cost: 0 % 5000 of 5000 (100%)

Clustered Index Seek (Cluste... [T2].[T2_PK] Cost: 0 % 0.0049

Clustered Index Seek (Cluste... [T3].[T3_PK] Cost: 0 % 0.0049

Query executed successfully. 302701845TFPC29 (17.0 RTM) 302701845TFPC29;stflap... QQ_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

SQLQuery1.s...aptop (55)*

```
--q6:
55 alter table T3 drop constraint T3_PK
56 alter table T3 add CONSTRAINT T3_PK primary key (b,a)
57
58
59 --q7:
60 -- Test 1: large value (returns 0 rows)
61 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
62 JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
63
64 -- Test 2: small value (returns all rows)
65 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
66 JOIN T3 on T1.a = T3.b WHERE T1.a > 0
67
68 -- Test 3: middle value
69 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
70 JOIN T3 on T1.a = T3.b WHERE T1.a > 2500
```

100 % No issues found Ln: 64, Ch: 1 (124 chars, 3 lines) SPC CRLF UTF-8

Results Messages Execution plan

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 0

SELECT

Cache plan size	80 KB
Estimated Operator Cost	0 (0%)
Degree of Parallelism	1
Estimated Subtree Cost	0.196421
Memory Grant	2752 KB
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	499.155

Statement

```
SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 0
```

Clustered Index Seek (Cluste...
[T1].[T3_PK]
Cost: 0 %
0.001a
5000 of
5000 (100%)

Clustered Index Seek (Cluste...
[T3].[T3_PK]
Cost: 0 %
0.001a

302701845TFPC29 (17.0 RTM) 302701845TFPC29\utflap... QQ_Fun_Quiz1 00:00:00 Row: 1, Col: 1 500 rows

Ready 61°F Mostly sunny Search 12:02 PM 4/29/2026

SQLQuery1.s...aptop (551)*

```

55 --q6:
56 alter table T3 drop constraint T3_PK
57 alter table T3 add CONSTRAINT T3_PK primary key (b,a)
58
59 --q7:
60 -- Test 1: large value (returns 0 rows)
61 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
62 JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
63
64 -- Test 2: small value (returns all rows)
65 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
66 JOIN T3 on T1.a = T3.b WHERE T1.a > 0
67
68 -- Test 3: middle value
69 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
70 JOIN T3 on T1.a = T3.b WHERE T1.a > 2500

```

100% No issues found Ln: 69, Ch: 1 (82 chars, 2 lines) SPC CRLF UTF-8

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 2500

Query executed successfully.

SQLQuery1.s...aptop (551)*

```

55 --q6:
56 alter table T3 drop constraint T3_PK
57 alter table T3 add CONSTRAINT T3_PK primary key (b,a)
58
59 --q7:
60 -- Test 1: large value (returns 0 rows)
61 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
62 JOIN T3 on T1.a = T3.b WHERE T1.a > 9999
63
64 -- Test 2: small value (returns all rows)
65 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
66 JOIN T3 on T1.a = T3.b WHERE T1.a > 0
67
68 -- Test 3: middle value
69 SELECT * FROM T1 JOIN T2 on T1.a = T2.a
70 JOIN T3 on T1.a = T3.b WHERE T1.a > 2500

```

100% No issues found Ln: 69, Ch: 1 (82 chars, 2 lines) SPC CRLF UTF-8

Query 1: Query cost (relative to the batch): 100%

SELECT * FROM T1 JOIN T2 on T1.a = T2.a JOIN T3 on T1.a = T3.b WHERE T1.a > 2500

Query executed successfully.

SELECT

Cached plan size	80 KB
Estimated Operator Cost	0 (0%)
Degree of Parallelism	1
Estimated Subtree Cost	0.137332
Memory Grant	2912 KB
Estimated Number of Rows for All Executions	0
Estimated Number of Rows Per Execution	248,294

Statement

SELECT * FROM T1 JOIN T2 on T1.a = T2.a
JOIN T3 on T1.a = T3.b WHERE T1.a > 2500